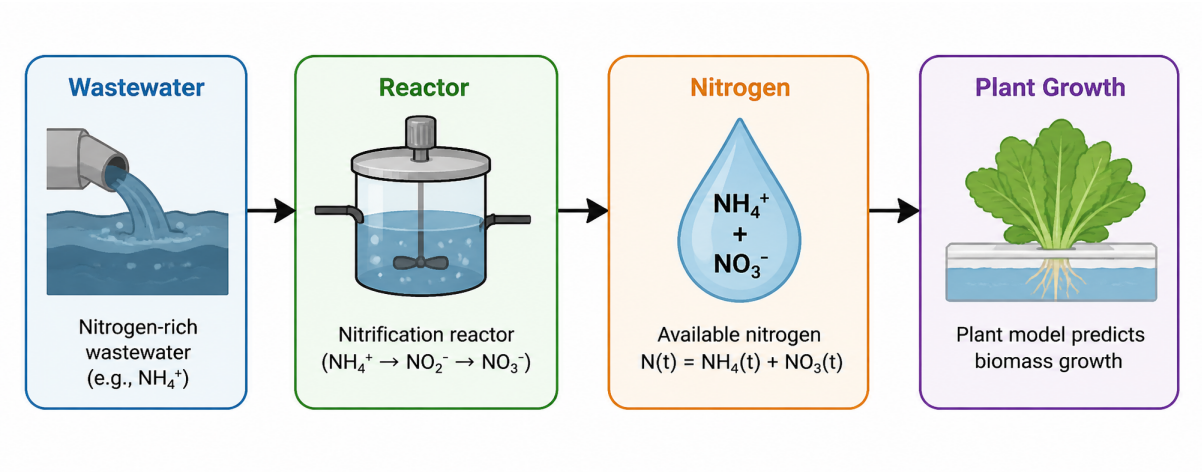

Simulating the Propagation of Nutrient Dynamics: A Coupled Simulation of Wastewater Nitrification, Nutrient Recovery, and Hydroponic Plant Growth

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Project GitHub Repository: `https://github.com/Jiasheng-Cao/CSE6730_Final`

1 Abstract

This project develops an integrated simulation framework to evaluate how wastewater-treatment reactor dynamics can propagate into downstream hydroponic crop growth through nutrient reuse. The system couples an upstream aerobic nitrification reactor with a downstream lettuce-growth module. The upstream model simulates ammonium, nitrite, nitrate, dissolved oxygen, ammonia-oxidizing bacteria, and nitrite-oxidizing bacteria in a continuous stirred-tank reactor. The downstream model uses plant-available nitrogen, defined as $N(t) = NH_4(t) + NO_3(t)$, as dynamic forcing for dry-mass prediction and nitrogen-uptake estimation. Five plant-growth formulations are evaluated: Dynamic-μ, RMND-μ, SMND-μ, Liebig minimum Monod, and multiplicative Monod. The integrated workflow compares an early usable-start coupling window with a later quasi-steady-start window, then matches the resulting nitrogen forcing to the closest experimental lettuce treatment. Both integrated scenarios match the R13 N-limited treatment. Validation of the quasi-steady Dynamic-μ simulation against 125 R13 dry-mass observations gives MAE = 0.1329 g, RMSE = 0.2934 g, and $R^2 = 0.7379$. Sensitivity analysis shows that final biomass is mainly controlled by nitrogen availability, while crop-cycle start timing has much weaker influence, especially under quasi-steady reactor conditions.

2 Project Description

The goal of this project is to develop a simulation framework that links wastewater treatment with hydroponic crop production, enabling the reuse of nutrients as a resource rather than treating them solely as waste. In conventional wastewater treatment systems, nutrients such as nitrogen are removed to meet discharge standards before releasing treated water into natural environments. While this approach protects water quality, it also discards nutrients that could otherwise be used to support plant growth. This project aims to explore an alternative perspective in which wastewater is treated as a source of recoverable nutrients for agriculture.

To achieve this goal, the system is modeled as two connected components: a wastewater treatment module and a hydroponic plant growth module. The wastewater treatment component represents an upstream biological reactor that transforms nitrogen in wastewater into different chemical forms. Specifically, ammonium is biologically converted into nitrite and then nitrate through a process known as nitrification. These transformations are carried out by specialized microorganisms, and the resulting nitrogen species differ in their suitability for plant uptake.

The downstream hydroponic module represents plant growth in a nutrient solution. In hydroponic systems, plants absorb nutrients directly from water rather than soil, making nutrient availability a key factor controlling growth. In this project, the concentrations of nitrogen species produced by the wastewater module are used as inputs to the plant growth model. This allows us to simulate how treated wastewater can support plant biomass production.

To capture the essential behavior of this integrated system, the model focuses on several key processes. In the wastewater module, the most important processes are biological nitrogen transformation, microbial growth, and oxygen availability. These determine how efficiently ammonium is converted into plant-available forms such as nitrate. In the plant growth module, the model focuses on how nutrient availability influences growth rate and biomass accumulation.

The system is simplified using several assumptions to maintain clarity and computational efficiency. The wastewater reactor is represented as a completely mixed system, meaning that concentrations are uniform throughout the reactor. Only aerobic nitrification is considered, while other nitrogen pathways are neglected. Dissolved oxygen is included as an important variable because it directly affects microbial activity and reaction rates. In the plant growth model, nutrient uptake is represented using simplified relationships that capture the dependence of growth on nutrient concentration.

By integrating these components, the model provides a framework for studying how wastewater-derived nutrients can be transformed and reused to support agricultural production. This ap-

proach enables analysis of the relationship between treatment conditions and plant growth, and provides insight into how wastewater treatment systems can be designed to support sustainable resource recovery.

3 Literature Review

Conventional wastewater treatment systems remove nutrients such as nitrogen and phosphorus to meet discharge standards before releasing effluent into natural water bodies. However, these nutrients could be reused for hydroponic crop production. Because wastewater models focus on nutrient removal and hydroponic studies treat nutrient solutions as fixed inputs, integrated mechanistic models linking treatment processes with hydroponic nutrient uptake and plant growth remain limited. Studies of integrated aquaponic simulation (i.e., fish-plant systems) provide examples of end-to-end module integration by explicitly representing nutrient generation and transformation in recirculating aquaculture systems along with subsequent plant-production processes[1,2]. At the municipal interface, system-level analysis has more often focused on nutrient recovery supply–demand assessment rather than dynamic treatment–cultivation integration[3]. Our project aims to develop a framework that enables system-level simulation of nutrient dynamics across the wastewater–hydroponic interface.

3.1 Mechanistic Modeling of Nitrogen Transformation in Activated Sludge Systems

The activated sludge process is the most widely applied biological wastewater treatment method worldwide, in which bacterial biomass (activated sludge) is responsible for removing pollutants such as excessive nitrogen from wastewater[4]. Various activated sludge configurations have been developed and implemented in wastewater treatment plants, and their corresponding mathematical modeling and simulation have long been essential tools for understanding and optimizing biological nitrogen removal under different operating conditions[5,6]. The ASM1 was the first standardized activated sludge model, followed by later extensions including ASM2, ASM2d, and ASM3[7]. Other widely used models such as Barker & Dold model[8] and the metabolic model developed at the Delft University of Technology (TUDP model)[9] have been published for researchers and wastewater treatment plant operators. Different ASM-based models vary in their targeted substrates and complexity, reflected in the number of processes, state variables, and parameters included[10]. However, all of these models simulate key nitrogen removal processes using a set of state variables and reaction equations based on mass-balance relationships and microbial kinetics[5]. For example, ASM1 describes the dynamic behavior of soluble and particulate substrates using mass-balance differential equations that account for influent flow, reactor volume, and biological reaction rates[6]:

$$\frac{dS_{liq,i}}{dt} = \frac{Q}{V_{liq}} \cdot (S_{in,i} - S_{liq,i}) + \sum_{j=1}^{19} \rho_j v_{i,j} \tag{1}$$

$$\frac{dX_{liq,i}}{dt} = \frac{Q}{V_{liq}} \cdot (X_{in,i} - X_{liq,i}) + \sum_{j=1}^{19} \rho_j v_{i,j} \tag{2}$$

Where: $S_{liq,i}$ is the concentration of each soluble state variable, $X_{liq,i}$ is the concentration of each particulate and biomass state variable, V_{liq} is the liquid reactor volume, Q is the flow in and out of the reactor, $S_{in,i}$ is the input concentration of soluble components, $X_{in,i}$ is the concentration of particulate and biomass components, $\sum_{j=1}^{19} \rho_j v_{i,j}$ is the sum of the specific kinetic rates ρ_j for process j multiplied by the stoichiometric coefficients $v_{i,j}$.

This project will be based on the existing ASM-based mechanistic modeling framework, but the simulation in this project will be designed to achieve controlled nitrogen transformation while retaining a portion of bioavailable nitrogen in the effluent for agricultural reuse. To

achieve this goal, selected state variables and kinetic parameters from traditional ASM models will be adjusted to represent partial nitrogen removal rather than full regulatory compliance. The resulting model will remain mechanistic but tailored to nutrient reuse.

3.2 Mechanistic Modeling and Experimental Studies of Hydroponic Cultivation and Reclaimed Wastewater Reuse

Mechanistic simulation of hydroponic plant systems has commonly focused on biomass dynamics together with nutrient uptake and accumulation, particularly nitrogen uptake and tissue nitrate accumulation[11,12]. These studies typically use state-variable ODE frameworks driven by environmental forcing and nutrient availability[11,12]. Representative examples include Michaelis–Menten uptake-based hydroponic dosing simulations validated on Brassica crops[13], as well as soilless greenhouse models for tomato, which update multi-ion solution composition using the uptake-concentration mass-balance approach for scenario and control analysis[14]. Similar kinetic simulation frameworks have also been applied to reclaimed-wastewater lettuce cultivation systems, in which saturating kinetics are used to represent nitrogen uptake and biomass productivity under variable nutrient supply[15].

Experimental studies of reclaimed wastewater nutrient reuse and pilot-scale operations have been more prevalent than integrated mechanistic simulations of wastewater–hydroponic systems. Reclaimed-water hydroponics has more often been investigated through integrated treatment–cultivation pilot systems that evaluate plant growth responses to treated municipal wastewater nutrient levels and typically do not extend to coupled mechanistic simulation[16].

Multi-nutrient growth limitation is commonly represented by Monod-type functions $f_i(S_i) = \frac{S_i}{K_{s,i}+S_i}$ combined across nutrients (e.g., multiplicative co-limitation $\mu = \mu_{max} \prod_i f_i$ or minimum/Liebig limitation), providing a standard starting point for N/P/K-resolved growth simulation[17]. Nutrient uptake and translocation are frequently represented with Michaelis–Menten influx $J_i = J_{max,i} \frac{C_i}{K_{m,i}+C_i}$ and embedded in solution mass balances; this approach has been implemented in hydroponic system simulators using multi-ion uptake with transpiration coupling[13] and in vertical lettuce nutrient film technology modeling where Michaelis–Menten uptake is paired with Monod growth to estimate uptake and biomass responses under nutrient variability[15]. Recent lettuce kinetic studies have extended the Monod multi-nutrient core by introducing Dynamic-μ (maturity-associated decline) with explicit N/P/K limitation, yielding a compact, simulation-ready cultivation kernel coupled to biomass via $\frac{dDM}{dt} = \mu$[18,19]. Complementary mechanistic frameworks for soilless culture formalize the coupling among nutrient and water uptake and root-zone transport processes[20], while controlled-environment lettuce modeling has further combined physics-based transpiration (Penman–Monteith) with structured nutrient-uptake submodels under closed-loop fertigation[21].

Together, these studies provide a foundation of process relationships and governing equations that can be adapted and integrated into physics-driven simulations of wastewater–hydroponic systems. Building on these established mechanistic formulations, the proposed framework seeks to simulate how wastewater-derived nutrient profiles influence hydroponic nutrient uptake, biomass dynamics, and overall system performance across the wastewater–agriculture interface.

4 Conceptual Model and Mathematical Framework

4.1 Mechanistic Modeling of Nitrogen Transformation

The nitrogen transformation process in the reactor was described using a mechanistic model that combines kinetic expressions for microbial growth and substrate utilization with mass-balance equations for soluble nitrogen species and biomass concentrations. In this framework, the kinetics define how ammonia-oxidizing bacteria (AOB) and nitrite-oxidizing bacteria (NOB) respond to substrate and environmental conditions, while the mass balances describe how ammonium,

nitrite, nitrate, dissolved oxygen, and microbial biomass change over time in the completely mixed reactor.

The growth of AOB and NOB was represented using Monod-type kinetic expressions. For AOB, the specific growth rate was defined as:

$$\mu_{AOB} = \mu_{AOB,\max}\left(\frac{S_{NH4}}{K_{NH4}+S_{NH4}}\right)\left(\frac{S_{O2}}{K_{O2,AOB}+S_{O2}}\right) \tag{3}$$

Similarly, the specific growth rate of NOB was defined as:

$$\mu_{NOB} = \mu_{NOB,\max}\left(\frac{S_{NO2}}{K_{NO2}+S_{NO2}}\right)\left(\frac{S_{O2}}{K_{O2,NOB}+S_{O2}}\right) \tag{4}$$

These expressions reflect that nitrifier activity depends on both substrate availability and dissolved oxygen concentration, and the Monod form allows the growth rate to increase with substrate concentration and asymptotically approach a maximum value.

To account for environmental effects, the maximum specific growth rates were further corrected for temperature and pH. Temperature dependence was described using an Arrhenius-type relationship:

$$\mu_{\max,T} = \mu_{\max,20} \cdot \theta^{(T-20)} \tag{5}$$

and pH dependence was incorporated as:

$$\mu_{\max,pH} = \mu_{\max} \cdot (1 - 0.833 \cdot (7.2 - pH)) \tag{6}$$

These correction terms allow the model to reflect the influence of operating conditions on nitrifying activity in a simplified manner.

The microbial growth rates were linked to nitrogen conversion through biomass yield coefficients. The ammonium utilization rate by AOB and the nitrite utilization rate by NOB were expressed as:

$$r_{AOB} = \frac{\mu_{AOB} X_{AOB}}{Y_{AOB}} \tag{7}$$

$$r_{NOB} = \frac{\mu_{NOB} X_{NOB}}{Y_{NOB}} \tag{8}$$

These relationships connect microbial growth with substrate removal, thereby linking biomass dynamics directly to nitrogen transformation.

To improve the engineering realism of the model, dissolved oxygen (DO) was incorporated as a dynamic state variable. Unlike the previous formulation where DO was treated as a constant, the updated model represents oxygen dynamics through a mass-balance equation that accounts for aeration-driven oxygen transfer and biological consumption by nitrifying microorganisms.

The DO balance was described as:

$$\frac{dS_{O_2}}{dt} = k_L a(S^*_{O_2} - S_{O_2}) - 3.43\, r_{AOB} - 1.14\, r_{NOB} \tag{9}$$

where $k_L a$ is the volumetric oxygen transfer coefficient, $S^*_{O_2}$ is the oxygen saturation concentration, and the last two terms represent oxygen consumption associated with ammonium and nitrite oxidation, respectively. In this study, $k_L a$ and $S^*_{O_2}$ were assumed constant, representing fixed aeration intensity and environmental conditions. Influent dissolved oxygen was neglected under the assumption that oxygen supply is dominated by aeration rather than influent flow. Additionally, oxygen consumption was assumed to arise solely from nitrification, while contributions from heterotrophic processes were not considered.

Based on these kinetic expressions, the system was described using a set of six coupled ordinary differential equations, representing the dynamic mass balances of three soluble nitrogen species, dissolved oxygen, and two key functional microbial groups. The nitrogen balances in the CSTR were written as:

$$\frac{dS_{NH4}}{dt} = \frac{Q}{V}(S_{NH4,in} - S_{NH4}) - r_{AOB} \tag{10}$$

$$\frac{dS_{NO2}}{dt} = \frac{Q}{V}(S_{NO2,in} - S_{NO2}) + r_{AOB} - r_{NOB} \tag{11}$$

$$\frac{dS_{NO3}}{dt} = \frac{Q}{V}(S_{NO3,in} - S_{NO3}) + r_{NOB} \tag{12}$$

The biomass balances for AOB and NOB were expressed as:

$$\frac{dX_{AOB}}{dt} = (\mu_{AOB} - b_{AOB})X_{AOB} - \frac{Q}{V}X_{AOB} \tag{13}$$

$$\frac{dX_{NOB}}{dt} = (\mu_{NOB} - b_{NOB})X_{NOB} - \frac{Q}{V}X_{NOB} \tag{14}$$

Together, these equations provide a dynamic mechanistic representation of two-step nitrification in a completely mixed reactor, allowing simultaneous simulation of nitrogen species, dissolved oxygen, and nitrifier biomass over time.

4.2 Mechanistic Modeling of Plant Growth

The plant growth module is based on a biomass-growth state equation in which nutrient availability determines the instantaneous specific growth rate, and the specific growth rate in turn governs biomass accumulation. Dry mass (DM) is used to represent biomass accumulation in plants. The biomass balance is written as

$$\frac{dDM}{dt} = \mu(t)\, DM \tag{15}$$

where $DM(t)$ denotes dry mass at time t, and $\mu(t)$ denotes the specific growth rate. This formulation represents biomass accumulation as an ordinary differential equation driven by the instantaneous growth rate and is consistent with mechanistic hydroponic cultivation models in which biomass evolution is governed by nutrient availability [11,12,15].

Nutrient limitation is represented using Monod-type saturation functions,

$$f_i(S_i) = \frac{S_i}{K_{s,i} + S_i} \tag{16}$$

where S_i is the available concentration of nutrient i, and $K_{s,i}$ is the half-saturation coefficient for nutrient i. This project focused on nitrogen, phosphorus, and potassium concentrations (N, P, and K), which are macro-nutrients for plant growth. These nutrients are supplied by the upstream wastewater-treatment module in the integrated framework.

The Liebig minimum Monod model is written as

$$\mu = \mu_{\max} \min\left(f_N(N), f_P(P), f_K(K)\right) \tag{17}$$

which assumes that growth is controlled by the single most limiting nutrient. The multiplicative Monod model is written as

$$\mu = \mu_{\max}\, f_N(N)\, f_P(P)\, f_K(K) \tag{18}$$

which assumes simultaneous co-limitation by all included nutrients.

The Dynamic-μ model is written as

$$\mu = \mu_{\max}\frac{N}{K_s + N} + m \max(0, t - \tau) \tag{19}$$

where $\mu_{\max}$ is the maximum specific growth-rate coefficient, K_s is the nitrogen half-saturation coefficient, m is the slope of maturity-associated decline after the delay point, and τ is the

delay time. This formulation extends the Monod-type nitrogen-limitation term by allowing the specific growth rate to decrease after a certain developmental stage. The RMND-μ model is a category-aware multi-nutrient formulation:

$$\mu = \mu_{\max,c}\, f_N(N)\, f_P(P)\, f_K(K) + m_c \max(0, t - \tau_c) \tag{20}$$

where subscript c indicates category-dependent parameters. The SMND-μ model uses a shared parameter set across retained conditions:

$$\mu = \mu_{\max}\, f_N(N)\, f_P(P)\, f_K(K) + m \max(0, t - \tau) \tag{21}$$

The Dynamic-μ formulation used here is based on published lettuce kinetic studies[18,19].

In addition to biomass accumulation, the downstream module also estimates nitrogen uptake from the upstream-derived nitrogen forcing. Nitrogen uptake is represented using a Michaelis–Menten saturation relationship:

$$U_N(t) = U_{\max,N} \frac{N(t)}{K_{m,N} + N(t)}$$

where $U_N(t)$ is the specific nitrogen uptake rate, $U_{\max,N}$ is the maximum nitrogen uptake rate, and $K_{m,N}$ is the half-saturation coefficient for nitrogen uptake. This formulation allows uptake to increase with nitrogen availability while approaching a maximum rate at high nitrogen concentration. This uptake expression provides an additional downstream response variable for estimating plant nitrogen demand, rather than serving as the fitting objective for dry-mass growth calibration.

For the Dynamic-based growth formulations, model parameters can be treated as cultivation-specific calibration variables, generically denoted by θ (e.g., including growth, half-saturation, and maturity-related terms as applicable to the selected model), rather than universal constants. These growth parameters are estimated by minimizing the mismatch between simulated dry-mass trajectories and measurements at the sampled growth days under the prescribed nutrient inputs for each treatment:

$$J(\theta) = \sum_j \left(DM_j^{obs} - DM_j^{pred}(\theta) \right)^2, \tag{22}$$

with the calibrated parameter set obtained as

$$\theta^* = \arg\min_\theta J(\theta). \tag{23}$$

5 Simulation Methodology

The simulation workflow was implemented in Python using NumPy, Pandas, and Matplotlib. The model consists of two connected stages: upstream nitrification simulation and downstream plant-growth simulation. The upstream module generates time-resolved nitrogen species concentrations, and the downstream module uses the reactor output as dynamic nutrient forcing for biomass prediction.

5.1 Upstream Nitrification Simulation

The upstream module simulates nitrification in a CSTR. The state variables are

$$S_{NH_4}, \quad S_{NO_2}, \quad S_{NO_3}, \quad S_{O_2}, \quad X_{AOB}, \quad X_{NOB},$$

where S_{NH_4}, S_{NO_2}, and S_{NO_3} are nitrogen species concentrations, S_{O_2} is DO, and X_{AOB} and X_{NOB} are nitrifier biomass concentrations. The upstream simulation is run over a 100-day operating period. The resulting time vector and solution matrix are imported into the integration workflow.

A mechanistic ODE model was used because nitrification depends on mass balances, microbial kinetics, DO limitation, and biomass washout. This method preserves the internal reactor states needed for downstream coupling. A simpler empirical curve-fit model could have been used, but it would not represent state-level nitrogen transformation. A full ASM-based model could also have been used, but it would require more state variables and parameters than were needed for the project scope.

5.2 Numerical Integration

The dynamic equations were solved using time-domain numerical integration. RK4 was used for the main ODE integration steps. RK4 was selected because it is explicit, deterministic, and sufficiently accurate for the smooth ODE systems used in this project. It also keeps the numerical method transparent and easy to verify.

Forward Euler was not selected because it is less accurate and more sensitive to time-step size. Adaptive solvers such as `solve_ivp` could have been used, but they would reduce implementation transparency and were not necessary for the current system. Implicit stiff solvers could be useful for larger wastewater models, but the simplified reactor and plant-growth formulations did not require that level of numerical complexity.

5.3 Downstream Plant-Growth and Nitrogen Uptake Simulation

5.3.1 Model Implementation

The downstream module simulates lettuce dry-mass accumulation and nitrogen uptake in response to a supplied nitrogen forcing signal. This plant-growth and nitrogen uptake module translates nutrient availability into time-resolved specific growth-rate, biomass, and uptake responses. In the current workflow, dry mass (DM) is used as the primary biomass state variable and is the simulated target variable of the plant-growth models.

To represent alternative ways of describing nutrient-limited plant growth, the downstream module includes five candidate formulations for the specific growth rate, μ: a nitrogen-focused Dynamic-μ model, a category-aware RMND-μ model, a shared SMND-μ model, a Liebig minimum Monod model, and a multiplicative Monod model. The Liebig minimum Monod and multiplicative Monod models are the most direct Monod-type growth models in the present module because they require only a maximum specific growth-rate parameter and nutrient half-saturation coefficients. In contrast, the Dynamic-μ, RMND-μ, and SMND-μ formulations include additional time-dependent or category-dependent parameters. Thus, while all five formulations use nutrient concentrations to compute $\mu(t)$, the simpler Monod-based models are more practical for scenario testing, whereas the Dynamic-μ-based formulations are most appropriate when experimental biomass data are available for calibration.

The plant-growth simulation first evaluates the nutrient-limitation functions and representative growth-rate expression for the selected model formulation. For each time point, the supplied nutrient forcing is used to compute the instantaneous specific growth rate $\mu(t)$. This growth-rate term is then incorporated into the biomass governing equation,

$$\frac{dDM}{dt} = \mu(t)DM.$$

To simulate the temporal evolution of dry mass, the biomass ODE is integrated over the cultivation period using the fourth-order Runge–Kutta method. This produces a time-resolved dry-mass trajectory that can be evaluated at the sampled growth days and compared with observed plant biomass data.

In addition to biomass simulation, the downstream module estimates nitrogen uptake using a Michaelis–Menten saturation relationship:

$$U_N(t) = U_{\max,N}\frac{N(t)}{K_{m,N} + N(t)},$$

where $U_N(t)$ is the specific nitrogen uptake rate, $U_{\max,N}$ is the maximum nitrogen uptake rate, $K_{m,N}$ is the half-saturation coefficient, and $N(t)$ is the supplied plant-available nitrogen concentration. This formulation allows nitrogen uptake to increase with nitrogen availability while approaching a maximum value at high nitrogen concentration. Including this uptake component allows the downstream module to evaluate plant nutrient demand in addition to dry-mass accumulation. More detailed plant models including light, transpiration, root-zone transport, or environmental controls could also be used, but these alternatives were not selected because the current implementation focuses on nutrient-driven growth and the required environmental inputs were not available in the dataset.

5.3.2 Experimental Data Usage and Parameter Fitting

The downstream implementation uses lettuce data collected from a vertical nutrient film technique (NFT) system operated with recovered nutrients from reclaimed wastewater under nutrient-limited conditions. These data are used for downstream model evaluation and for calibration of the Dynamic-μ-based growth-model parameters. Therefore, the dataset serves not merely as a case-specific validation source, but as the experimental basis for establishing calibrated parameter sets that represent lettuce growth behavior under the modeled reclaimed-water hydroponic conditions. These calibrated parameter sets are reused in the integrated framework, where upstream-predicted nitrogen concentrations are supplied as dynamic nutrient inputs to the downstream plant-growth and nitrogen-uptake simulations.

The original data used in this study are openly available in the USDA Ag Data Commons repository as *Dataset for the Hydroponic Cultivation of Bibb Lettuce in NPK-Limited Conditions*, DOI: `https://doi.org/10.15482/USDA.ADC/28801286.v1`, accessed on 22 August 2025. For the present workflow, the fitting data are taken from the `ModSummary` sheet, whereas the plotting and validation data are taken from the `ModPlant` sheet. Treatment-level nutrient values for N, P, and K are extracted from the dataset and used as forcing inputs to the cultivation simulations. Although the full downstream dataset and analysis include N-, P-, K-limited, and MSS treatments (nutrient-sufficient control), the present report focuses primarily on the N-limited cases in order to align with the upstream nitrogen-transformation analysis.

The workflow loads the lettuce NFT dry-mass data, removes excluded rows such as invalid growth stages or screening rows, fills nutrient inputs using sample-period or lifetime-average values, assigns treatment categories, and aggregates observations by treatment and growth day. Constant treatment-level N, P, and K values are then passed into the downstream simulation as forcing inputs for model calibration, downstream module validation, and treatment-level evaluation. These treatment-level forcing inputs also provide the basis for matching upstream-generated nitrogen conditions to previously calibrated and validated lettuce treatments in the integrated simulation.

Parameter fitting is performed using dry-mass measurements for the Dynamic-μ-based formulations. For Dynamic-μ, RMND-μ, and SMND-μ, the corresponding parameter vector is estimated by minimizing the difference between observed and simulated dry mass at the sampled growth days. The same overall optimization structure is used for these formulations, with the parameter vector adjusted according to the selected growth-rate model.

The nitrogen uptake component is parameterized separately from the dry-mass growth-model calibration. Its Michaelis–Menten parameters describe uptake-rate behavior and are used to estimate nitrogen uptake from the supplied nitrogen forcing, while the Dynamic-μ-based growth-model calibration is based on matching simulated and observed dry-mass trajectories.

5.3.3 Validation and Visualization Workflow

After calibration, the fitted Dynamic-μ-based parameter sets are used to simulate treatment-level dry-mass trajectories and to compare predicted values against observed dry-mass measurements at the sampled growth days. The Liebig minimum Monod and multiplicative Monod formulations are retained as simpler Monod-type baseline models and are evaluated under the

same nutrient-forcing conditions for comparison. For each retained treatment, the model is initialized using the observed early dry mass, integrated over the cultivation period, and evaluated at the same growth days as the observations. Model performance is then quantified using standard regression metrics, including R^2, mean absolute error (MAE), and root mean square error (RMSE).

For visualization, the candidate growth formulations are overlaid within the same treatment panel so that differences among model structures can be inspected directly. This allows the downstream module to compare how the Dynamic-μ, RMND-μ, SMND-μ, Liebig minimum Monod, and multiplicative Monod formulations represent dry-mass growth under the same treatment-level nutrient conditions. The resulting calibrated parameter values, dry-mass simulations, validation metrics, and comparative plots provide the downstream model basis used in the integrated workflow.The main downstream outputs from this module are the specific growth-rate trajectory $\mu(t)$, the dry-mass trajectory $DM(t)$, and the nitrogen-uptake trajectory $U_N(t)$. The supplied nitrogen signal is treated as the input to the downstream module and is defined in the coupling step described in Section 5.4. These outputs allow the downstream module to translate a supplied nitrogen condition into plant biomass accumulation and nutrient-demand responses.

5.4 Upstream–Downstream Coupling and Integrated Validation Design

5.4.1 Coupling Workflow

The upstream–downstream coupling step converts the reactor output into the nitrogen forcing used by the downstream plant-growth and uptake module. The coupling variable is plant-available nitrogen, defined as

$$N(t) = S_{NH_4}(t) + S_{NO_3}(t),$$

where $S_{NH_4}(t)$ and $S_{NO_3}(t)$ are expressed on a nitrogen-mass basis (mg N L^{-1}). Therefore, ammonium and nitrate concentrations can be summed directly without molecular-weight conversion. Nitrite is retained in the upstream reactor simulation but is not used as a downstream plant-available nitrogen input because it is usually a transient nitrification intermediate rather than a major nitrogen source for lettuce growth. This choice keeps the coupling focused on the dominant inorganic nitrogen forms used by plants and avoids introducing additional assumptions about nitrite uptake.

Phosphorus and potassium are treated as non-limiting constants in the integrated simulation. This assumption isolates the nitrogen pathway, which is the main coupling variable between the reactor and plant-growth modules in the current project. More complete downstream nutrient coupling could include dynamic P and K availability, but this was not implemented because the present integration focuses on nitrogen transformation from the upstream nitrification reactor.

The crop-cycle duration is inferred from the experimental plant-growth dataset as

$$T_{crop} = 32 \text{ d}.$$

The upstream reactor output is mapped onto the crop-growth time domain by selecting a coupling start time t_s and extracting the upstream forcing over

$$[t_s,\ t_s + T_{crop}].$$

The downstream crop time is therefore defined relative to the selected start time, so that reactor operating time is reparameterized into crop-growth time.

Linear interpolation is used to evaluate the upstream nitrogen signal on the downstream simulation grid. Linear interpolation was selected because the upstream reactor output is generated as a dense and smooth time series from the RK4 simulation. Higher-order interpolation was not used because it could introduce artificial oscillations or overshoot in the nutrient forcing and was not needed for the current time resolution.

Two start-window definitions are implemented to represent different upstream operating conditions. The usable-start window selects the earliest time after 5 days for which dissolved oxygen remains above 7.5 mg/L for a 3-day hold period:

$$S_{O_2}(t) \geq 7.5 \text{ mg/L} \quad \text{for} \quad t \in [t_s,\ t_s + 3].$$

The quasi-steady-start window selects the earliest time after 5 days for which plant-available nitrogen remains stable over a 10-day forward window:

$$\max_{\tau \in [t_s, t_s+10]} |N(\tau) - N(t_s)| \leq 0.05 \text{ mg N/L}.$$

These two windows were chosen to compare an early usable reactor condition with a later stable reactor condition. This method can distinguish transient operation from quasi-steady operation and provide insight into how temporary start-up dynamics and stabilized reactor output affect downstream biomass and uptake responses.

For each selected coupling window, the mean plant-available nitrogen concentration over the crop-cycle window is used to identify the closest previously calibrated N-limited treatment condition:

$$\overline{N}_{cycle} = \frac{1}{T_{crop}} \int_{t_s}^{t_s+T_{crop}} N(t)\, dt.$$

The corresponding downstream growth-parameter set and nitrogen-uptake parameterization from the matched treatment are then applied in the downstream simulation. This treatment-matching step provides a practical way to connect the continuous upstream nitrogen output with experimentally characterized lettuce-growth and nitrogen-uptake behavior.

The current integration is one-way coupled. The upstream reactor output affects downstream growth and uptake, but plant uptake does not feed back into the reactor or reduce the supplied nitrogen concentration. This structure was selected because the current project focuses on how reactor dynamics propagate into downstream plant-growth and nutrient-demand responses. A two-way coupled system would be more complete, but it would require additional assumptions about hydroponic solution volume, recirculation, nutrient depletion, and feedback from plant uptake to solution concentration.

The final integrated outputs from the coupled workflow are

$$N(t), \quad \mu(t), \quad DM(t), \quad U_N(t).$$

Together, these outputs allow the simulator to evaluate how upstream nitrification dynamics propagate into downstream plant biomass accumulation and nitrogen uptake demand.

5.4.2 Integrated Module Validation Design

The integrated module is evaluated after the upstream nitrification module and the downstream plant-growth module have been assessed individually. This creates three validation levels in the overall workflow: upstream validation against measured effluent nitrogen species, downstream validation against treatment-level lettuce dry-mass observations under experimental nutrient forcing, and integrated validation against plant-level dry-mass observations from the treatment matched to the upstream-generated nitrogen profile. This layered validation structure helps distinguish errors associated with the reactor model, the plant-growth model, and the coupled upstream–downstream workflow.

For integrated validation, the upstream-derived plant-available nitrogen trajectory is first mapped onto the crop-growth period using the selected coupling window. The mean plant-available nitrogen concentration over that crop-cycle window is then matched to the closest calibrated and validated N-limited lettuce treatment. The corresponding downstream growth-parameter set and uptake parameterization are then used to simulate dry-mass accumulation and nitrogen uptake under the dynamic upstream forcing. The quasi-steady-start case is used for detailed validation because it represents the more stable upstream operating condition and

provides a nitrogen forcing level that aligns with the lettuce treatment data. Integrated dry-mass predictions are compared with observed plant-level dry-mass data from the matched treatment using R^2, MAE, and RMSE. Bootstrap resampling is used to estimate confidence intervals for these validation metrics. The nitrogen uptake trajectory is also compared with observed treatment-level uptake summaries to evaluate whether the simulated uptake response is directionally consistent with the experimental data.

6 Experimental Results and Validation

6.1 Experimental Results and Validation of Nitrogen Transformation in the Wastewater Treatment Module

6.1.1 Simulation Design and Experimental Procedure

To evaluate the performance of the developed nitrification model, a total of 22 validation cases were simulated using experimentally measured influent ammonium concentrations as model inputs. These cases were obtained under aerobic nitrification conditions, in which ammonium was biologically oxidized to nitrite and nitrate.

For each case, the influent ammonium concentration was used as the primary input variable, while influent nitrite and nitrate were assumed negligible. All other model parameters, including kinetic constants, oxygen transfer coefficient (k_La), and environmental correction factors (temperature and pH), were kept constant across simulations. This approach isolates the effect of influent ammonium concentration and ensures consistency across validation scenarios.

Each simulation was run for 100 days to reach steady-state conditions. Effluent concentrations of ammonium (NH_4^+), nitrite (NO_2^-), and nitrate (NO_3^-) were then extracted and compared with experimental measurements. This comparison served as the basis for model validation.

6.1.2 Validation Approach and Output Analysis

Model validation was performed by comparing simulated and experimental effluent concentrations across all 22 cases. The prediction error for each case was defined as:

$$e_i = C_{sim,i} - C_{exp,i} \tag{24}$$

Based on these errors, statistical metrics including mean error, mean absolute error (MAE), root mean square error (RMSE), and 95% confidence intervals were calculated for each nitrogen species. Table 1 summarizes the statistical results.

Table 1: Error metrics for effluent nitrogen species across the validation dataset

Species	Mean Error (mg N/L)	95% CI (mg N/L)	MAE (mg N/L)	RMSE (mg N/L)	R^2
NH_4^+	0.677	[-0.003, 1.357]	1.317	1.644	-0.204
NO_2^-	0.545	[-0.012, 1.103]	1.247	1.344	-0.197
NO_3^-	-1.873	[-3.032, -0.714]	2.630	3.167	0.019

The results indicate systematic bias in model predictions. For ammonium (NH_4^+), the model consistently overestimates residual concentrations, with a mean error of +0.68 mg N/L and a 95% confidence interval of [-0.003, 1.36] mg N/L. The RMSE is 1.64 mg N/L. For nitrite (NO_2^-), the model also shows overestimation, with a mean error of +0.55 mg N/L, a 95% confidence interval of [-0.01, 1.10] mg N/L, and an RMSE of 1.34 mg N/L.

For nitrate (NO_3^-), the model exhibits a consistent negative bias, with a mean error of -1.87 mg N/L and a 95% confidence interval of [-3.03, -0.71] mg N/L. The RMSE for nitrate is 3.17 mg N/L, indicating the largest deviation among the three species. The fact that the confidence interval for nitrate does not include zero confirms that this underestimation is systematic.

A parity plot (Figure 1) was used to further assess model performance. Ideally, all points would lie along the 1:1 line. The results show that nitrate predictions generally follow the experimental trend but consistently fall below the 1:1 line. In contrast, ammonium and nitrite predictions cluster within a narrow simulated range, indicating limited sensitivity of the model to variations in influent conditions.

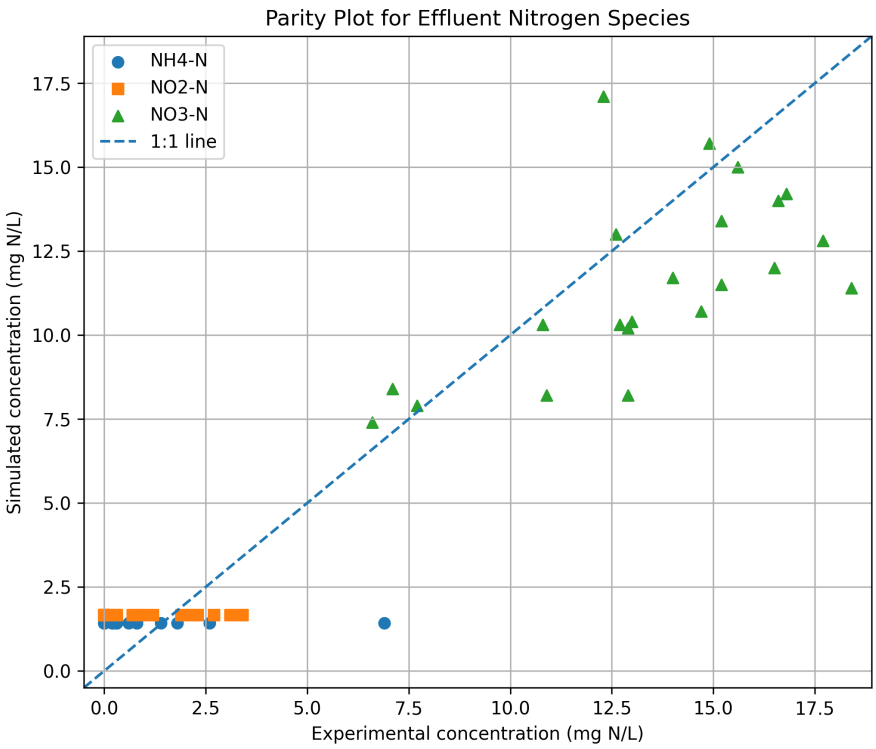

Figure 1: Parity plot comparing simulated and experimental effluent nitrogen concentrations. The dashed line represents perfect agreement.

6.1.3 Interpretation of Model Performance

Overall, the model successfully captures the dominant nitrification pathway, including the conversion of ammonium to nitrate and the presence of intermediate nitrite. However, the analysis reveals that the model does not adequately reproduce case-specific variability.

The relatively narrow confidence intervals combined with consistent bias indicate that the discrepancies are systematic rather than random. In particular, the clustering of simulated ammonium and nitrite values suggests that the model outputs are weakly responsive to changes in influent conditions. This behavior is consistent with the negative R^2 values observed for ammonium and nitrite, which indicate that the model fails to capture variability rather than overall trends.

These results suggest that the current model structure is sufficient to represent the general nitrification process but requires improved parameterization or additional mechanisms to capture dynamic variability. Potential sources of discrepancy include simplified kinetics, constant oxygen transfer assumptions, and the absence of inhibition or competing biological processes.

Despite these limitations, the model provides a mechanistic framework capable of describing nitrogen transformation and serves as a suitable foundation for further development of the integrated wastewater–hydroponic system.

6.2 Downstream Plant-Growth Module Validation

6.2.1 Validation Procedure and Output Analysis

The downstream plant-growth module was first evaluated independently from the upstream reactor module using treatment-level experimental nutrient inputs. In this downstream-only validation, constant treatment-level N, P, and K concentrations from the lettuce NFT dataset were used as forcing inputs, and the simulated dry-mass trajectories were compared with observed dry-mass measurements at the sampled growth days. This validation step was performed before integrated coupling so that the performance of the plant-growth module could be assessed separately from errors introduced by the upstream nitrification model or the coupling procedure.

The validation focused on N-limited treatments to align with the nitrogen-centered upstream transformation model. For each retained treatment, the downstream simulator was initialized using the observed early dry mass, simulated over the cultivation period, and evaluated at the same growth days as the experimental observations. Model performance was quantified using coefficient of determination (R^2), mean absolute error (MAE), and root mean square error (RMSE). Comparative trajectory plots were also generated to evaluate how different model structures represented dry-mass accumulation across the N-limited treatments.

6.2.2 Downstream Validation Results

Table 2 summarizes R^2, MAE, and RMSE for the N-limited downstream validation, together with bootstrap 95% confidence intervals for each metric. Bootstrap resampling with 5000 iterations was used to estimate the confidence intervals. The Dynamic-μ formulation produced the strongest performance in this N-limited validation subset, with $R^2 = 0.7679$, MAE = 0.4389 g, and RMSE = 0.9849 g. The bootstrap 95% confidence intervals were [0.7087, 0.8440] for R^2, [0.2401, 0.6811] g for MAE, and [0.4764, 1.4685] g for RMSE. RMND-μ and SMND-μ produced identical performance in the current implementation, while the Liebig minimum Monod and multiplicative Monod models showed lower aggregate R^2 and higher RMSE values. The relatively wide R^2 intervals for several formulations reflect treatment-level heterogeneity within the N-limited validation subset.

Table 2: Downstream dry-mass validation metrics for N-limited treatments with bootstrap 95% confidence intervals.

Model	R^2	R^2 95% CI	MAE	MAE 95% CI	RMSE	RMSE 95% CI
Dynamic-μ	0.7679	[0.7087, 0.8440]	0.4389	[0.2401, 0.6811]	0.9849	[0.4764, 1.4685]
RMND-μ	0.6862	[0.0861, 0.8666]	0.5509	[0.3170, 0.8250]	1.1452	[0.6355, 1.6147]
SMND-μ	0.6862	[0.0861, 0.8666]	0.5509	[0.3170, 0.8250]	1.1452	[0.6355, 1.6147]
Liebig minimum Monod	0.6573	[0.0186, 0.8891]	0.5254	[0.2755, 0.8273]	1.1967	[0.6203, 1.7223]
Multiplicative Monod	0.6333	[-0.0697, 0.8801]	0.5399	[0.2825, 0.8516]	1.2380	[0.6271, 1.7988]

Although the aggregate metrics provide an overall ranking, treatment-level behavior remained heterogeneous. Figure 2 presents the treatment-level dry-mass trajectories for the N-limited cases, comparing the observed data with the fitted simulations. The Dynamic-μ model provided the best aggregate agreement across the N-limited validation subset, but treatment-level differences were still evident. RMND-μ and SMND-μ produced identical aggregate metrics in the current implementation and showed similar behavior across the evaluated N-limited treatments. The Liebig minimum Monod and multiplicative Monod models captured the general increase in biomass with nitrogen availability, but their lower R^2 values and higher RMSE values indicate weaker overall agreement with the observed N-limited dry-mass trajectories. The lower confidence bound for the multiplicative Monod R^2 was slightly negative, indicating that some bootstrap-resampled subsets were poorly represented by this formulation.

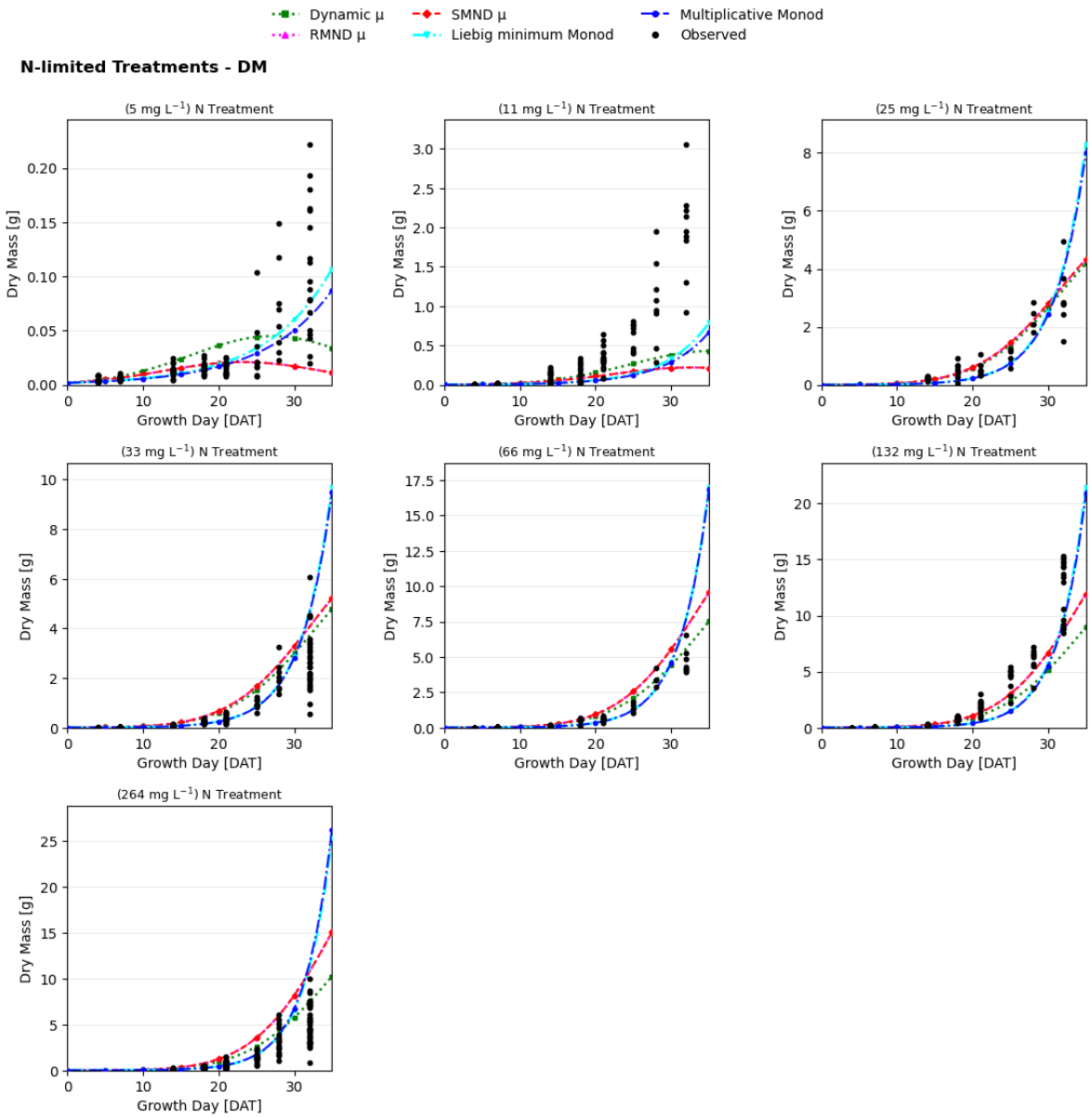

Figure 2: N-limited Plant Growth: observed and simulated dry-mass trajectories under N-limited treatments using the Dynamic-μ and Liebig minimum Monod models.

6.2.3 Interpretation of Downstream Module Validation

The downstream validation results show that the plant-growth module can reproduce the general dry-mass accumulation patterns observed under reclaimed-wastewater nutrient-limited hydroponic conditions. In the N-limited validation subset, Dynamic-μ provided the strongest aggregate agreement with observed dry-mass data, indicating that the nitrogen-focused Dynamic-μ formulation was effective for representing biomass growth when nitrogen limitation was the primary focus. RMND-μ and SMND-μ produced identical metrics in the current implementation, so the distinction between these two multi-nutrient Dynamic-μ-based formulations should be interpreted cautiously and may require further examination of the parameterization workflow.The baseline Monod-type formulations provided useful comparison cases but showed weaker aggregate performance than Dynamic-μ in the N-limited validation. The Liebig minimum Monod model performed slightly better than the multiplicative Monod model, suggesting that the minimum-limitation assumption may better represent the present N-limited downstream dataset than fully multiplicative co-limitation. However, the wide bootstrap confidence intervals, especially for R^2, indicate that treatment-level variability remains important and that model ranking

should be interpreted within the scope of the N-limited validation subset.The downstream-only validation should be interpreted separately from the integrated module validation. In this section, nutrient forcing is based on experimental treatment-level N, P, and K inputs, so the results evaluate the plant-growth simulator under known nutrient conditions. In contrast, the integrated validation in the next section evaluates whether upstream-generated nitrogen trajectories can be propagated through the coupling workflow to produce realistic downstream dry-mass and uptake responses. Therefore, Section 6.2 establishes the standalone performance of the downstream plant-growth module under N-limited treatment-level forcing, while Section 6.3 evaluates the coupled upstream–downstream system.

6.3 Integrated Model Simulation Results and Validation

6.3.1 Integrated upstream–downstream coupling results

The integrated simulation links the upstream nitrification reactor output to the downstream hydroponic plant-growth module by using time-resolved plant-available nitrogen as the external forcing signal. The upstream output is then mapped onto a 32-day crop-growth period by selecting a crop-cycle start time and interpolating the upstream nitrogen signal onto the downstream simulation grid. Phosphorus and potassium are treated as non-limiting constants so that the integrated analysis focuses on the propagation of upstream nitrogen dynamics into plant dry-mass growth.

Two upstream operating windows were selected to represent different coupling conditions. The usable-start window begins at 5.00 d, which corresponds to the earliest post-startup period satisfying the dissolved-oxygen operability criterion. The quasi-steady-start window begins at 35.94 d, which corresponds to the earliest period where total plant-available nitrogen remains approximately stable over a 10-day forward window. These two windows allow the integrated framework to compare an early operable reactor condition with a later nitrogen-stable condition. As shown in Figure 3, the usable-start window captures a more transient portion of the upstream nitrogen profile, whereas the quasi-steady-start window is located after the upstream nitrogen signal has become smoother and more stable.

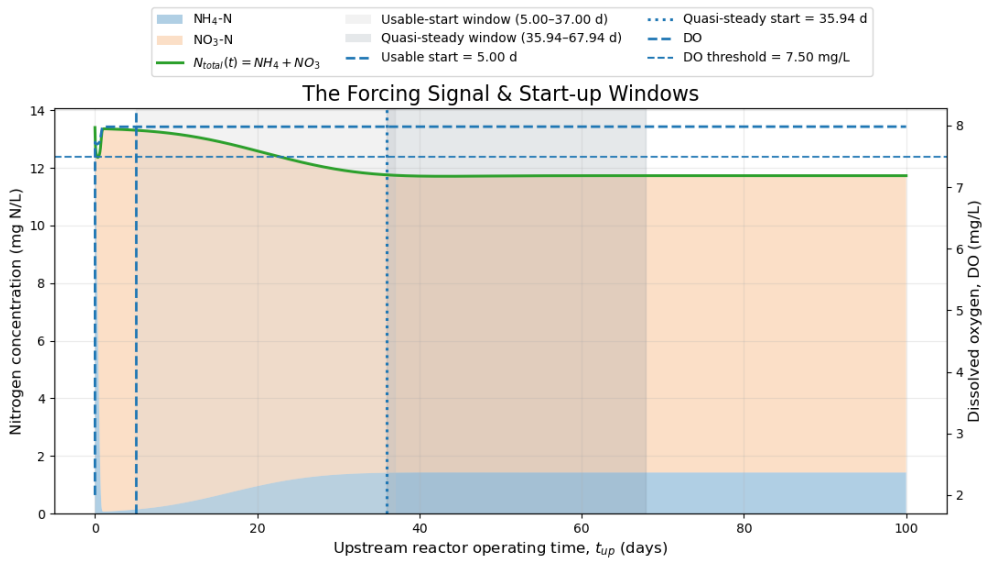

Figure 3: The Forcing Signal & Start-up Windows

For each selected crop-cycle window, the mean upstream nitrogen level was matched to the closest experimental treatment input concentration from the calibrated plant-growth dataset. Both the usable-start and quasi-steady-start cycles were matched to treatment R13, an N-limited treatment with an input nitrogen target of 11 mg N/L. The usable-start cycle had a

mean total nitrogen concentration of approximately 12.52 mg N/L, while the quasi-steady-start cycle had a mean total nitrogen concentration of approximately 11.72 mg N/L. Therefore, both integrated simulations use the same matched treatment category, downstream growth-parameter set, and nitrogen-uptake parameterization, allowing the effect of upstream operating-window selection to be isolated.

The Dynamic-μ model was first used to compare the usable-start and quasi-steady-start coupling scenarios. The usable-start case produced a slightly higher nitrogen forcing level during the crop cycle and therefore yielded a slightly higher growth-rate trajectory and final dry mass. In contrast, the quasi-steady-start case produced a more stable nitrogen input profile and a smoother downstream response. This comparison shows that the integrated framework can transmit differences in upstream reactor operating conditions into downstream growth-rate and biomass trajectories, even when the downstream model structure and matched treatment are held fixed.

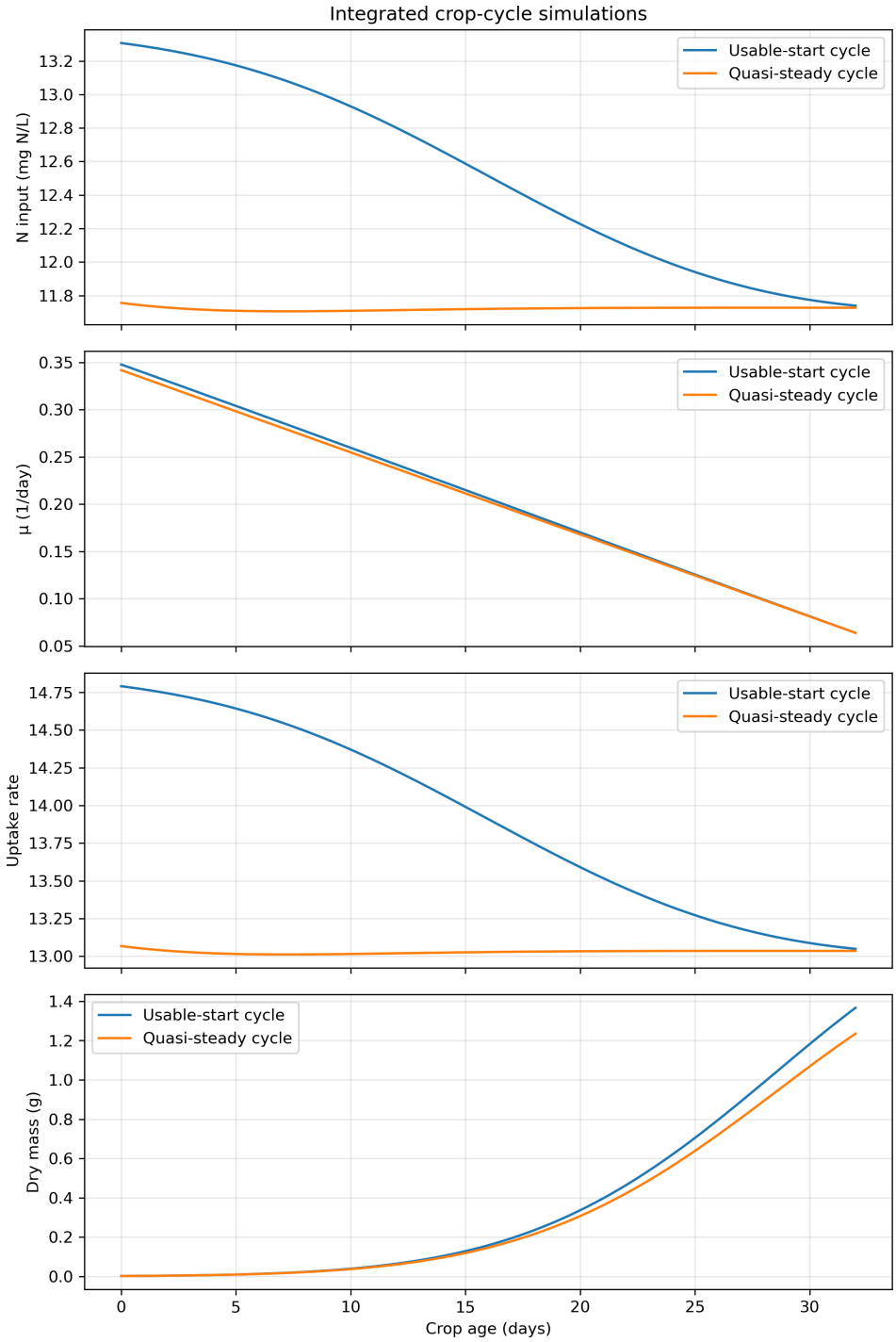

Figure 4: Integrated crop-cycle simulations

6.3.2 Comparison across growth formulations and validation against observations

To evaluate the effect of downstream model structure, the integrated nitrogen forcing was applied to five candidate growth formulations: Dynamic-μ, RMND-μ, SMND-μ, Liebig minimum Monod, and multiplicative Monod. All five models were evaluated under the same upstream-driven nitrogen input, while phosphorus and potassium were kept non-limiting. This comparison separates uncertainty due to upstream forcing from uncertainty due to the downstream growth formulation.

The multi-model integrated comparison shows that the predicted dry-mass trajectories differ noticeably across growth formulations. Under the same nitrogen forcing, the Liebig and multiplicative Monod formulations tend to produce larger final dry mass because their nutrient-

limitation structure does not include the same maturity-associated decline term used in the Dynamic-μ-based formulations. In contrast, the Dynamic-μ, RMND-μ, and SMND-μ models produce more conservative late-stage growth because their fitted growth-rate expressions include time-dependent decline. This difference indicates that downstream model structure is an important source of uncertainty in the integrated system, even when the upstream nitrogen signal is fixed.

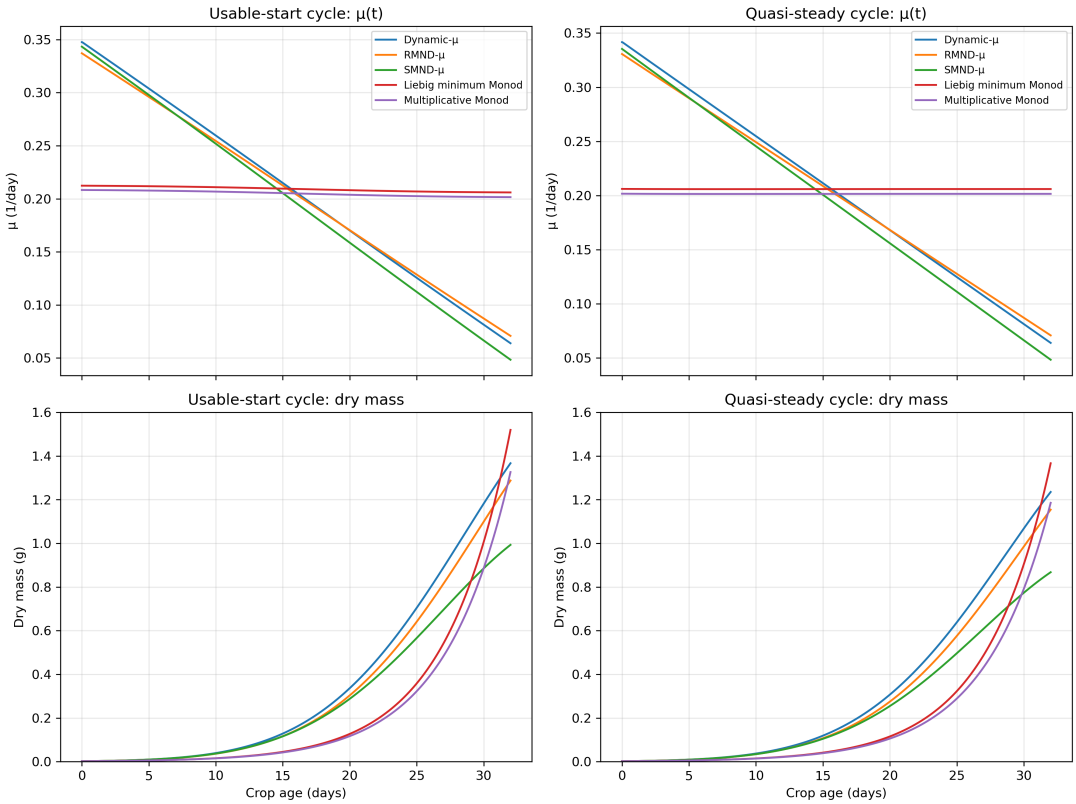

Figure 5: Integrated growth-model comparison under upstream-driven nitrogen forcing

The quasi-steady-start Dynamic-μ simulation was then validated against dry-mass observations from the matched R13 treatment. This validation used 125 plant-level observations from the experimental dataset. The resulting error metrics were MAE = 0.1329 g, RMSE = 0.2934 g, and $R^2 = 0.7379$. Bootstrap resampling with 5000 iterations gave 95% confidence intervals of [0.0921, 0.1843] for MAE, [0.1894, 0.3978] for RMSE, and [0.6670, 0.8167] for R^2. These results indicate that the integrated quasi-steady Dynamic-μ simulation captures the observed R13 dry-mass trend reasonably well, with a positive R^2 and relatively small absolute prediction errors for the matched treatment.

Table 3: Validation metrics for the quasi-steady-start Dynamic-μ integrated simulation against R13 dry-mass observations.

Metric	Estimate	95% CI lower	95% CI upper
MAE (g)	0.1329	0.0921	0.1843
RMSE (g)	0.2934	0.1894	0.3978
R^2	0.7379	0.6670	0.8167

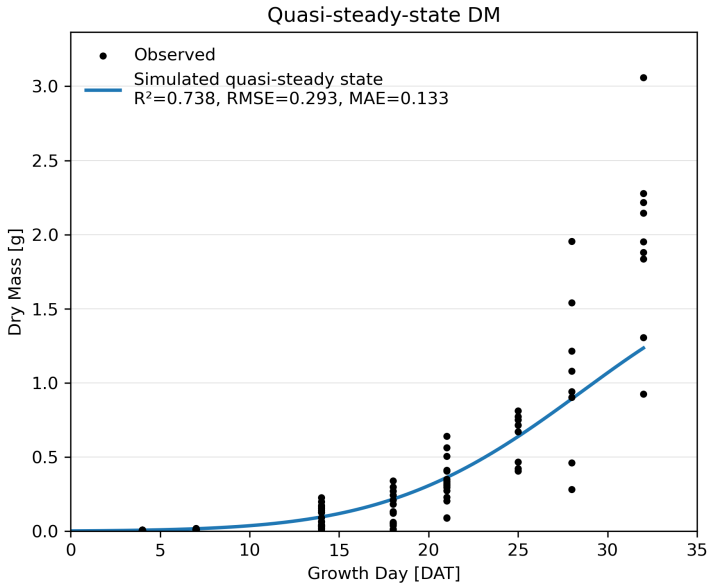

Figure 6: Quasi-steady-state DM

A multi-model validation was also performed for the quasi-steady-start cycle. In this matched-treatment validation, the Dynamic-μ model provided the best agreement with the R13 observations among the tested integrated formulations, with the lowest MAE and RMSE and the highest R^2. This result differs from the broader downstream-only validation in Section 6.2, where model performance was evaluated across multiple treatments under constant experimental nutrient inputs. The integrated validation here is narrower because it focuses on the upstream-driven quasi-steady R13 scenario. Therefore, the result should be interpreted as evidence that Dynamic-μ is the most suitable formulation for this specific integrated R13 coupling case, rather than as a universal ranking across all nutrient treatments.

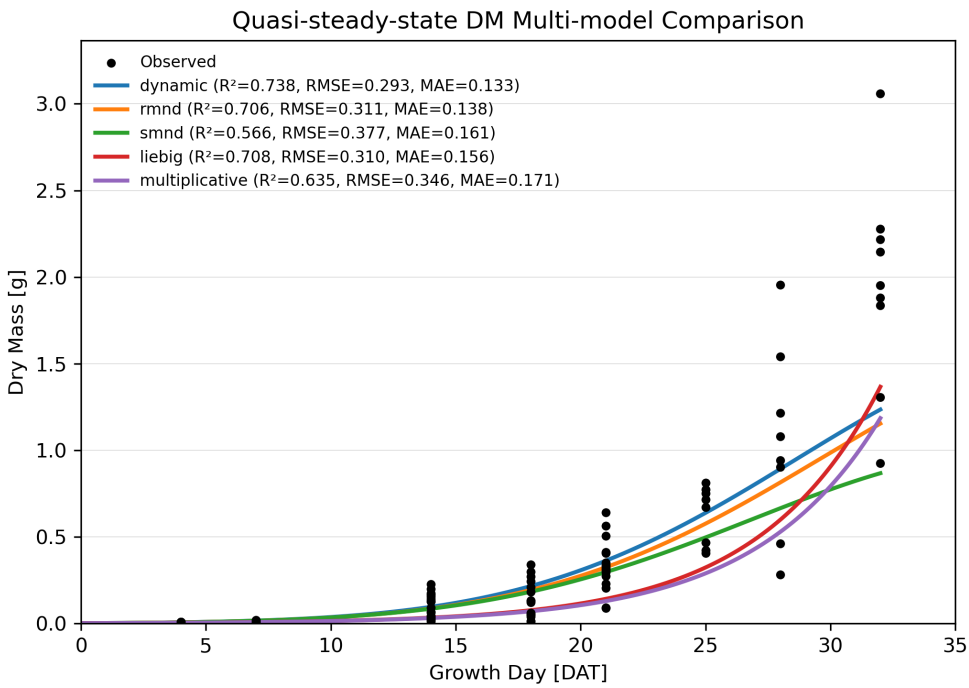

Figure 7: Quasi-steady-state DM Multi-model Comparison

6.3.3 Nitrogen uptake diagnostic

In addition to dry-mass prediction, the integrated model was used to estimate nitrogen uptake using the fitted Michaelis–Menten uptake relationship. For the quasi-steady-start R13 scenario, the simulated nitrogen uptake trajectory had a mean value of 13.03 mg nutrient g^{-1} dry mass d^{-1}, with a narrow range from 13.01 to 13.07 mg nutrient g^{-1} dry mass d^{-1}. The corresponding observed R13 nitrogen uptake averages were 10.81 mg nutrient g^{-1} dry mass d^{-1} at the sample/rig level and 10.52 mg nutrient g^{-1} dry mass d^{-1} at the treatment/day level. Thus, the simulated uptake is somewhat higher than the observed mean but remains in the same order of magnitude. This suggests that the uptake component is directionally consistent with the experimental data, although future work should improve the uptake calibration and include feedback from plant uptake to nutrient depletion.

6.3.4 Sensitivity analysis

Sensitivity analysis was then conducted to determine whether the integrated model output is more strongly controlled by nitrogen-forcing magnitude or by crop-cycle start timing. Two perturbation variables were tested. First, the upstream nitrogen forcing $N(t)$ was multiplied by scaling factors of 0.70, 0.85, 1.00, 1.15, and 1.30. This perturbation represents uncertainty or operational variation in the amount of plant-available nitrogen supplied by the upstream reactor. Second, the crop-cycle start time was shifted by −5, −2, 0, 2, and 5 days relative to the selected usable-start and quasi-steady-start windows. This perturbation tests whether the integrated output is sensitive to the exact timing of downstream coupling.

Figure 8 shows the one-factor response of final dry mass to nitrogen-forcing scale. For both usable-start and quasi-steady-start cycles, final dry mass increases consistently as the nitrogen scaling factor increases from 0.70 to 1.30. This trend is observed across all five growth formulations, indicating that nitrogen availability is a dominant driver of downstream biomass production in the integrated system. The models differ in their absolute final biomass predictions, with the Liebig minimum Monod model generally producing the highest final dry mass and the SMND-μ model producing the lowest. However, the positive response to nitrogen scaling is consistent across model structures. This indicates that the sensitivity to nitrogen availability is not an artifact of a single growth formulation, but a shared behavior of the integrated framework.

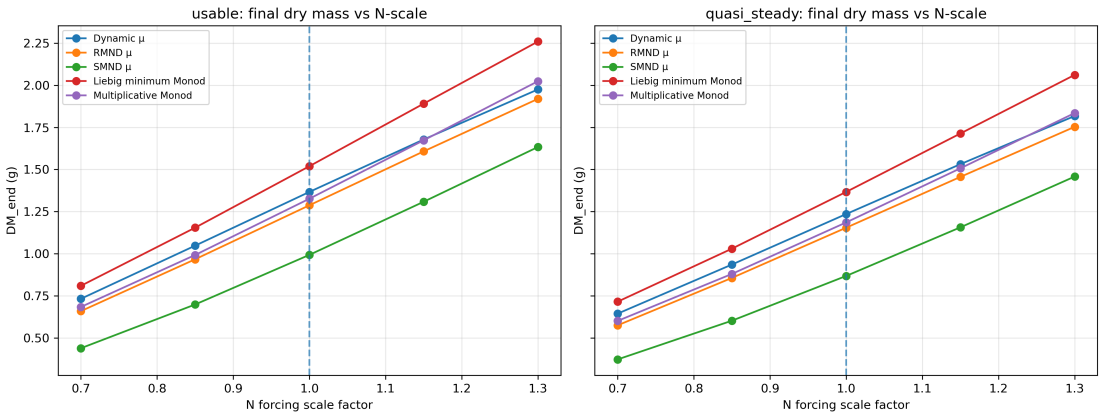

Figure 8: Final dry mass response to nitrogen forcing scale

Figure 9 shows the corresponding one-factor response to crop-cycle start-time shifts. Compared with the nitrogen-scaling response, the start-shift response is much weaker. Under the usable-start condition, shifting the crop cycle later causes a small decrease in final dry mass for all models. This occurs because the usable-start window is located in a more transient portion of the upstream reactor trajectory, where the nitrogen signal is still changing over time.

Therefore, moving the downstream crop cycle forward or backward slightly changes the nitrogen history experienced by the plant-growth module. In contrast, under the quasi-steady-start condition, the curves are almost flat for all five models. This indicates that once the upstream reactor output reaches a stable nitrogen regime, the integrated dry-mass prediction becomes nearly insensitive to small changes in crop-cycle timing.

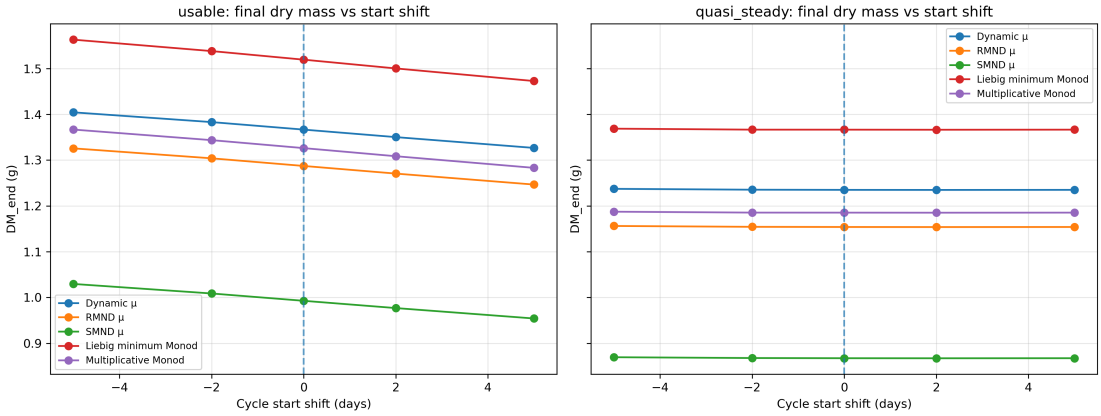

Figure 9: Final dry mass response to crop-cycle start-time shift

To evaluate the combined effect of the two perturbation variables, a full-factorial two-dimensional sensitivity analysis was also performed using nitrogen scaling and start-time shifting simultaneously. Figure 10 presents the percent change in final biomass relative to the baseline case. The heatmaps confirm the one-factor results. Changes along the nitrogen-scale direction are much larger than changes along the start-shift direction, producing a strong horizontal gradient in final biomass response. In the usable-start case, start-time shifts introduce some additional variation because the selected window overlaps with a transient upstream period. In the quasi-steady-start case, the heatmap is dominated almost entirely by nitrogen scaling, while the start-shift effect is minimal. This result reinforces that the integrated model is mainly sensitive to the magnitude of plant-available nitrogen rather than to small timing changes after the reactor has stabilized.

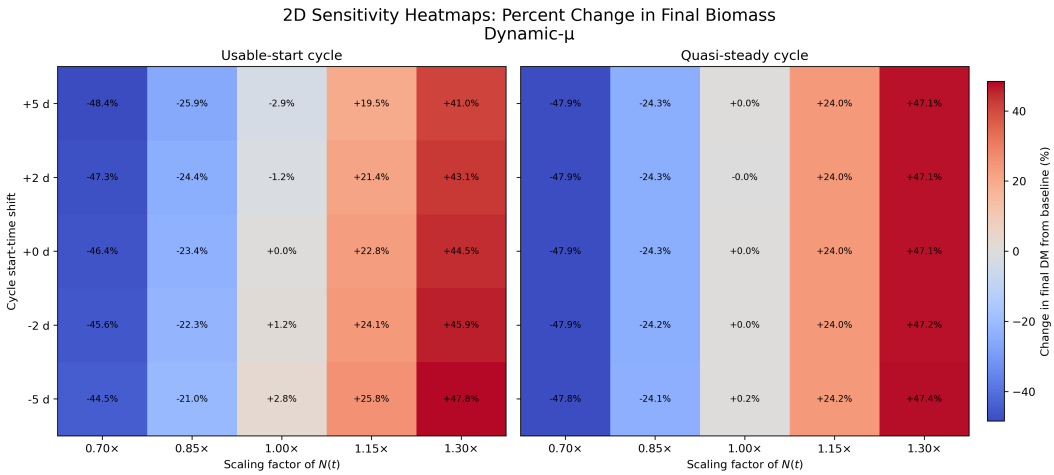

Figure 10: 2D sensitivity heatmaps: percent change in final biomass

The main-effect comparison further summarizes the relative importance of the two perturbation variables. For both usable-start and quasi-steady-start cycles, the range in final dry mass caused by nitrogen scaling is much larger than the range caused by start-time shifts. This difference is especially strong under the quasi-steady-start condition, where the upstream nitrogen profile is already stable and downstream growth becomes nearly insensitive to small

timing perturbations. Therefore, the sensitivity analysis supports a clear system-level conclusion: the integrated model is primarily resource-driven rather than timing-driven. In other words, model outputs depend much more on the amount of available nitrogen supplied by the upstream reactor than on small variations in the crop-cycle start date.

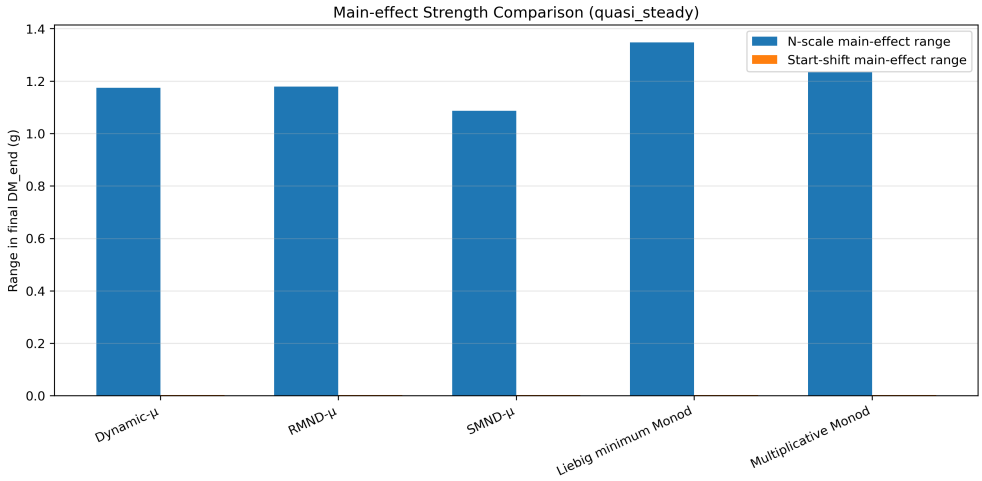

Figure 11: Main-effect strength comparison for the quasi-steady-start cycle

Overall, the nitrogen uptake diagnostic and sensitivity analysis show that the coupled framework can translate upstream nitrogen availability into downstream growth and uptake behavior. The uptake prediction is somewhat higher than the observed R13 uptake average but remains within the same order of magnitude, suggesting that the uptake module is directionally reasonable. The sensitivity results further show that final biomass is mainly controlled by nitrogen forcing magnitude, while start-time uncertainty becomes relatively unimportant once the upstream reactor reaches quasi-steady operation. The current one-way coupling is therefore useful for evaluating how upstream reactor operation affects downstream crop growth. However, future work should improve the framework by adding two-way feedback between plant uptake and nutrient concentration, recalibrating uptake under dynamic nutrient conditions, and expanding validation beyond the R13 matched treatment.

7 Discussion and Conclusion

7.1 Key Insights from the Modeled System

This project developed a coupled simulation framework linking an upstream nitrification reactor with a downstream hydroponic plant growth system, enabling evaluation of how nitrogen transformation in wastewater treatment influences plant growth.

From the wastewater treatment module, the model successfully captured the dominant nitrification pathway, including ammonium oxidation, transient nitrite formation, and nitrate production. However, validation against experimental data revealed systematic discrepancies. Ammonium and nitrite concentrations were generally overestimated, while nitrate was underestimated. These deviations were consistent across all validation cases, indicating that the model captures overall trends but has limited ability to reproduce case-specific variability. This behavior is further reflected by the negative R^2 values for ammonium and nitrite, which suggest that the model outputs are weakly sensitive to changes in influent conditions rather than indicating complete model failure.

From the hydroponic module, the simulations showed that model structure strongly affects predicted biomass growth. Different growth formulations produced different trajectories under the same nitrogen input, with Dynamic-μ type models providing more realistic behavior under nutrient-limited conditions compared to simple Monod-type models.

From the integrated system perspective, a key insight is that nitrogen availability is the primary driver of plant growth. Sensitivity analysis showed that variations in nitrogen concentration significantly affect final biomass, while variations in crop-cycle start timing have relatively minor effects, especially under quasi-steady reactor conditions. This indicates that the system is primarily resource-driven rather than timing-driven, and that stable upstream nitrogen output leads to more predictable downstream performance.

Overall, the study demonstrates that even a simplified mechanistic model can effectively link wastewater nitrogen transformation to plant growth, while also highlighting limitations in capturing variability.

7.2 Recommendations for Future Work

Several improvements can be made to extend this work and improve the modeling framework.

First, the upstream nitrification model can be refined by including additional processes such as inhibition effects (e.g., free ammonia and free nitrous acid), heterotrophic activity, and variable oxygen transfer conditions. These improvements would likely reduce the systematic bias observed in the validation results and increase model sensitivity to changing inputs.

Second, the hydroponic growth model can be expanded by incorporating additional nutrients (e.g., phosphorus and potassium) and environmental factors such as light and temperature. This would allow the model to better represent realistic plant growth conditions.

Third, the current system uses one-way coupling, where wastewater outputs determine plant growth, but plant uptake does not influence nutrient concentrations. Future work could introduce two-way coupling to represent feedback between plant uptake and nutrient availability, especially in recirculating systems.

Finally, model validation can be strengthened by using a broader range of experimental datasets and operating conditions. This would improve the robustness of the model and increase confidence in its predictive capability.

Overall, the current framework provides a solid foundation for modeling wastewater–hydroponic integration, and future improvements should focus on enhancing dynamic responsiveness and system realism.

8 Division of labor

Jiasheng Cao and Zijun Meng mainly worked on the first part of the simulation framework: the upstream nitrification simulation. They collaborated on refining and optimizing the reactor model, which simulates the transformation of nitrogen species in wastewater. Their work focused on improving the numerical stability and performance of the reactor simulation, validating the nitrogen transformation behavior using previous experimental data, and preparing the reactor outputs for downstream coupling with the plant growth model.

Yuming Sun was responsible for developing the downstream hydroponic plant-growth and nitrogen-uptake module and implementing the upstream–downstream integration framework. This included constructing the dry-mass and nitrogen-uptake simulations, processing the lettuce growth dataset for model calibration and downstream validation, evaluating multiple plant-growth formulations under varying nutrient conditions, mapping upstream-generated nitrogen trajectories to the crop-growth period, comparing usable-start and quasi-steady-start coupling scenarios, and performing integrated model validation using matched plant-level dry-mass observations. She also contributed to the corresponding written sections, visualizations, and analysis for the downstream and integrated modeling components.

Building on this integrated framework, Ruixin Liu conducted sensitivity analysis to evaluate the system's response to key factors, including nitrogen availability and cycle start time. This work also explored potential directions for further model improvement and optimization by comparing different growth formulations and identifying dominant system drivers. Ruixin Liu was responsible for the sensitivity analysis of the integrated model, including testing the effects

of nitrogen-forcing scale and crop-cycle start-time shifts on final dry mass. She also calculated the confidence intervals for the integrated model validation metrics, contributed to the overall report outline and selected sections of the written report, and conducted visualization analysis to support the interpretation of model behavior and key findings.

ChatGPT was used as an auxiliary tool during this project to support code development for selected visualizations and partial metric calculations. Specifically, AI assistance was used to help refine plotting code and compute selected validation metrics. All model design choices, result interpretation, report writing, and final conclusions were written and reviewed by the team.

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